Introduction to Causal Inference

Stage 2: Project Research Proposal

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1 Causal Question

"Does sustained daily cigarette smoking throughout pregnancy causally increase the probability of infant death before age one?"

Treatment Variable

- **Variable names:** `CIG1_R`, `CIG2_R`, `CIG3_R` (daily cigarette count recodes per trimester; values: 0 = no cigarettes per day, 1 = 1–5/day, ..., 5 = ≥41/day, 6 = unknown).
- **Treated group ($T = 1$):** Mother smoked at least one cigarette per day on average in *every* trimester (`CIG1_R` ≥ 1, `CIG2_R` ≥ 1, `CIG3_R` ≥ 1), ensuring sustained exposure throughout the entire pregnancy.
- **Control group ($T = 0$):** Mother reported zero cigarettes per day in every trimester (`CIG1_R` = 0, `CIG2_R` = 0, `CIG3_R` = 0).

Outcome Variable

- **Definition:** $Y = 1$ if the infant died before their first birthday, $Y = 0$ otherwise; derived from `FLGND` (death-linkage indicator), appended to every birth record in the NCHS denominator-plus file [7].

Temporal ordering. Smoking during pregnancy is recorded on the birth certificate before the infant is born, while infant death can only occur after birth. Reverse causality is therefore impossible by design.

Primary estimand. Using the potential outcomes framework, let $Y^{(1)}$ denote infant mortality under sustained smoking and $Y^{(0)}$ under never-smoking. Our primary estimand is the Average Treatment Effect (ATE), $\tau = E[Y^{(1)} - Y^{(0)}]$. We will also report the Average Treatment Effect on the Treated (ATT) for methods such as propensity score matching, which naturally focus on treated mothers.

2 Existing Knowledge and Related Literature

The relationship between prenatal smoking and adverse infant outcomes is among the most studied topics in perinatal epidemiology. Dietz et al. [1] estimate that roughly 5–8% of preterm births, 13–19% of term low-birth-weight deliveries, and 5–7% of infant deaths in the United States are attributable to prenatal smoking, placing the population-attributable fraction at roughly one in fourteen infant deaths. Cnattingius [2] documents that smoking prevalence during pregnancy is strongly socioeconomically patterned, which motivates our concern about confounding.

Biologically, nicotine causes uteroplacental vasoconstriction, carbon monoxide displaces fetal oxygen, and tobacco impairs placental angiogenesis, converging on hypoxia, growth restriction, preterm delivery, and abruption [2]. Wisborg et al. [3] found a dose-dependent association with SIDS (adjusted OR ≈3.5); Salihu and Wilson [4] document heterogeneous effects by dose and mortality subtype.

Despite this large observational literature, few studies apply formal causal methods. Most adjust for birth weight which is a post-treatment mediator. Our contribution is to bring an explicit causal framework to a question largely studied through associational methods

3 Data

We use the **2015 Cohort Linked Birth/Infant Death Data Set** [7], published by the National Center for Health Statistics (NCHS), CDC. The file follows all U.S. infants born in 2015 through their first birthday, covering approximately 3.98 million births of which 23,357 are linked to a death certificate (99.4% linkage rate). Birth certificate data, recording maternal characteristics, prenatal behaviour, and obstetric information, are collected at delivery and submitted by states to NCHS under the Vital Statistics Cooperative Program. When an infant dies, the corresponding death certificate is filed and linked back to the birth record by the state of death, yielding a single combined record per infant. The outcome $Y = 1$ is assigned to infants with a linked death record, $Y = 0$ otherwise.

Key **observed pre-treatment confounders** available in the data:

Variable	**Field**	**Role**
Mother's education	`MEDUC`	SES proxy; predicts smoking and infant health
Mother's age	`MAGER`	Younger mothers smoke more; age affects mortality risk
Race/Hispanic origin	`MRACEHISP`	Differential smoking prevalence and healthcare access
Marital status	`DMAR`	Social support and economic resource proxy
WIC participation	`WIC`	SES proxy; WIC confers nutrition and care benefits
Payment/insurance	`PAY_REC`	Medicaid vs. private insurance as SES indicator
Pre-pregnancy diabetes	`RF_PDIAB`	Pre-existing risk factor for adverse outcomes
Pre-pregnancy hypert.	`RF_PHYPE`	Pre-existing risk factor
Pre-pregnancy BMI	`BMI_R`	Reflects pre-existing health status before pregnancy
Father's education	`FEDUC`	Household SES proxy correlated with maternal smoking
Father's age	`FAGE11`	Household SES proxy correlated with maternal smoking

Mediators. Birth weight (`BRTHWGT`) and gestational age (`COMBGEST`) are **post-treatment mediators**: the literature consistently shows that smoking causes low birth weight and prematurity, which in turn raise infant mortality risk [1, 2].

4 Causal Assumptions

4.1 Consistency

Smoking is self-reported on the birth certificate at delivery, which introduces some risk of misreporting, mothers may underreport smoking due to social desirability. However, it is unlikely that a mother who smoked daily throughout every trimester would report zero cigarettes, nor that a true non-smoker would report otherwise. Our binary threshold therefore preserves the clinically meaningful contrast between sustained daily exposure and complete non-smoking.

4.2 Stable Unit Treatment Value Assumption (SUTVA)

We think that interference is implausible: there is no conceivable mechanism by which one mother's smoking behaviour during pregnancy could affect another infant's mortality risk. Regarding treatment versions, although mothers differ in the number of cigarettes smoked per day, the treatment is well-defined and binarised, sustained daily smoking versus complete non-smoking, which abstracts away from dose heterogeneity.

4.3 Conditional ignorability

We assume that, conditional on the observed covariates, treatment assignment is independent of the potential outcomes. The dataset is rich for a population-level study, covering socioeconomic status, maternal health history, and demographics, and several variables, education, insurance, WIC participation, serve as partial proxies for harder-to-measure constructs. Nevertheless, the literature consistently highlights maternal mental health, psychological stress, alcohol consumption, and drug use as linked to both smoking persistence and adverse infant outcomes [2, 8], none of which are recorded in birth certificate data. To the extent these unobserved factors remain associated with both treatment and outcome, which may be the case, for example if mothers facing greater psychological stress tend to sustain smoking more and also face higher infant mortality risk, our estimates may be upward biased. That said, since these unobserved factors likely correlate with observed controls, conditioning on the latter partially absorbs their confounding influence, limiting residual bias.

4.4 Positivity

After applying all filters, the analytic sample contains 2,938,232 records: 174,225 sustained smokers (5.9%) and 2,764,007 never-smokers (94.1%). We estimated the propensity score $e(\mathbf{X}) = P(T = 1 \mid \mathbf{X})$ via logistic regression on all eleven confounders. The propensity score distributions overlap substantially between treated and controls, so positivity holds nominally.

5 Pearl's Causal Graph

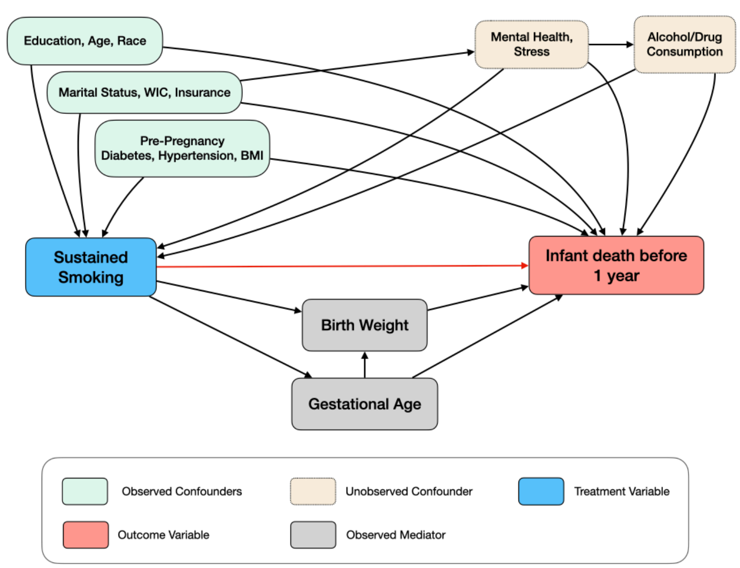

Figure 1: Causal graph encoding the assumed causal structure linking sustained maternal smoking (T) to infant death before age one (Y).

6 Anticipated Challenges

Residual confounding. The primary threat to identification comes from unobserved confounders, such as maternal mental health, stress, and substance use, that may affect both smoking persistence and infant mortality. These variables open unblocked backdoor paths between treatment and outcome in the causal graph. As a result, the graph as drawn does not formally support identification of the ATE by adjustment on the observed covariates alone. We therefore interpret our estimates as causal only under the strong and untestable assumption that residual confounding from these unobserved factors is negligible.

Practical positivity violations. Although the propensity score distributions overlap substantially, a large share of controls have propensity scores near zero. This near-zero mass creates highly unstable inverse probability weights, which may inflate variance and render IPW estimators unreliable. We address this by trimming observations outside the 1st and 99th percentiles of the treated propensity score distribution as a pre-specified sensitivity check.

Self-reported treatment. Smoking is self-reported at delivery, and mothers may under-report their smoking behavior due to social desirability. An extreme case would be a sustained smoker declaring zero cigarettes across all trimesters, which would misclassify her into the control group and bias estimates downward. We assume such complete misreporting is not systematic, it is unlikely that a large share of daily smokers would consistently report zero cigarettes throughout an entire pregnancy, but we acknowledge it as a plausible concern that cannot be fully ruled out.

Rare outcome. Infant death affects approximately 0.6% of the sample, which makes outcome modelling challenging: parametric models may be more sensitive to overfitting or quasi-complete separation. This concern is partly mitigated by the very large sample size, especially for matching-based methods, where the large control pool increases the chance of finding comparable never-smoking mothers for treated cases. Nevertheless we will still monitor model generalization carefully and consider regularization where needed.

7 Planned Estimation Methods

We deliberately apply four estimators rather than committing to a single approach, so as to provide a spectrum of estimates that can be jointly assessed for consistency and robustness.

$\mathbf{X}$ is the set of observed confounders while birth weight and gestational age are excluded from all adjustment sets, as they are post-treatment mediators.

Outcome regression. We will regress Y (infant death indicator) on T (sustained smoking indicator) and $\mathbf{X}$, using both a linear probability model and logistic regression. We will implement two outcome-regression approaches. First, we will use an S-learner, fitting a single outcome model on T and $\mathbf{X}$ jointly. The treatment effect will be estimated by comparing, for each observation, the predicted outcome under sustained smoking with the predicted outcome under never-smoking, and then averaging these differences over the sample.

Second, we will use a T-learner, fitting separate outcome models for treated and control units. The treatment effect will then be estimated by comparing the predicted outcomes from the treated and control models for the same covariate profiles, and averaging these differences across observations.

Both estimators target the ATE under the working assumption of conditional ignorability given the observed covariates. However, as discussed, this assumption is not guaranteed by the study design, since some relevant pre-treatment common causes are unobserved. We therefore interpret these estimates as causal only under the additional assumption that residual confounding is negligible.

Propensity score matching (PSM). We model $P(T = 1 \mid \mathbf{X})$ via logistic regression with T as the label and $\mathbf{X}$ as covariates, then match each of the 174,225 treated mothers to a nearest-neighbour control. The ATT is estimated as the mean difference in Y within matched pairs.

Inverse probability weighting (IPW). Using the same propensity score model, we will construct inverse-probability weights to create a pseudo-population in which treatment assignment is balanced with respect to the observed covariates. The ATE will then be estimated as the difference in weighted mean outcomes between the sustained-smoking and never-smoking groups. We may use stabilized weights and trim the tails to guard against extreme propensity scores.

Augmented IPW (AIPW). AIPW combines propensity score model above with two outcome models, $\hat{E}[Y \mid T = 1, \mathbf{X}]$ and $\hat{E}[Y \mid T = 0, \mathbf{X}]$, fit separately on each treatment arm. The augmentation term corrects for misspecification in either model, yielding a doubly robust ATE estimate.

8 Planned Sensitivity Checks

Because our estimates rest on modelling choices and untestable assumptions, we complement our main results with sensitivity checks designed to assess whether conclusions hold under alternative specifications.

Bootstrap confidence intervals. All confidence intervals will be constructed via nonparametric bootstrapping. Each bootstrap replicate will re-run the full estimation pipeline, outcome modelling, propensity score estimation, matching, and treatment-effect estimation, so that the resulting intervals propagate uncertainty from every step.

Trimming rules for IPW. Extreme propensity scores, concentrated near zero among controls (Section 6), can inflate IPW weights and destabilize estimates. We will compare three trimming rules: excluding observations outside the 1st–99th percentiles of the treated propensity score distribution, the 5th–95th percentiles, and a symmetric trim at the 10th–90th percentiles. Stability of the IPW estimate across these rules will indicate whether the main result is driven by a small set of poorly-supported observations.

Consistency across estimators. We will systematically compare point estimates and confidence intervals across all four methods (outcome regression, PSM, IPW, AIPW). Convergence across estimators will strengthen confidence in the result. Divergence, on the other hand, may reveal robustness issues and raise doubts about the validity of the underlying assumptions.

Propensity score calibration. The validity of PSM and IPW hinges on a well-specified propensity score model. We will assess calibration via the Brier score and a calibration curve (predicted versus observed treatment rates across deciles of $\hat{e}(\mathbf{X})$). Poor calibration would call into question the reliability of any weight- or match-based estimate.

Sensitivity to implementation choices. Several implementation decisions are inherently discretionary. For PSM, we will compare nearest-neighbor matching under Euclidean, Manhattan, and Minkowski distances on the logit-transformed propensity score to verify that matched-sample balance and the ATT estimate are not sensitive to the choice of metric. For the S-learner, T-learner, IPW, and AIPW estimators, we will tune the main hyperparameters of the underlying predictive models and report estimates across a range of configurations, so that the reader can assess whether results are an artefact of a particular tuning choice.

References

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Appendix

.1 Propensity Score Overlap Diagnostics

The following figures are produced by fitting a logistic regression of T on all eleven observed confounders (Table 1) using the analytic sample of 2,938,232 records. Figures are used to visually assess the overlap (positivity) assumption discussed in §4.4.

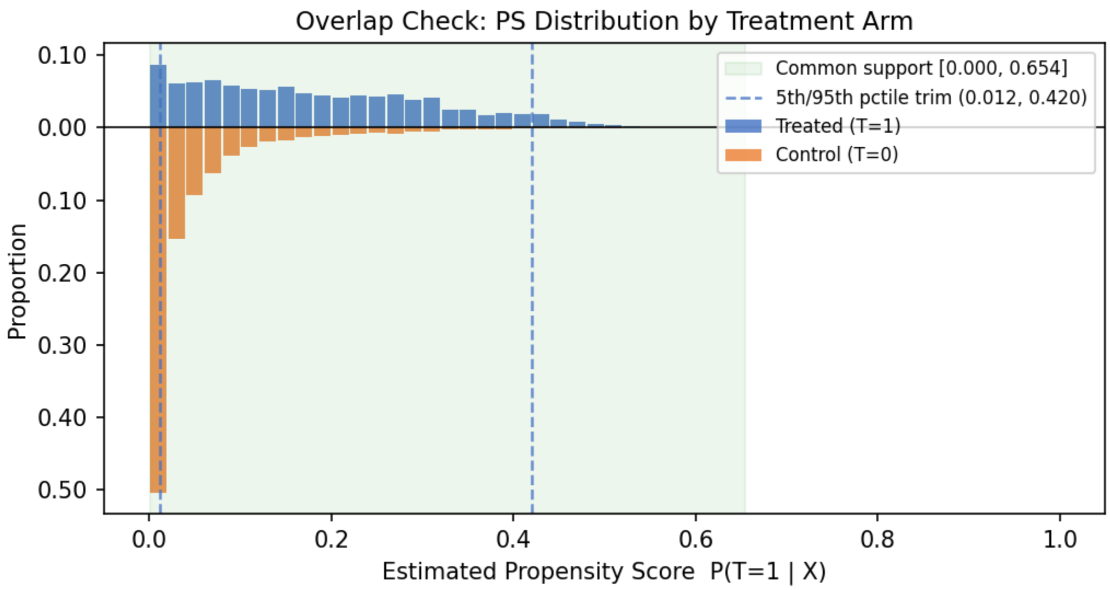

Figure 2: Mirror histogram of estimated propensity scores $\hat{e}(\mathbf{X})$ by treatment arm. Treated units (blue, above axis) are compared with control units (orange, below axis), both normalised to proportions. The shaded green region marks the common support [0.0003, 0.654]; dashed lines indicate the 5th and 95th percentiles of the treated PS (0.012 and 0.420) used for the trimming sensitivity check.

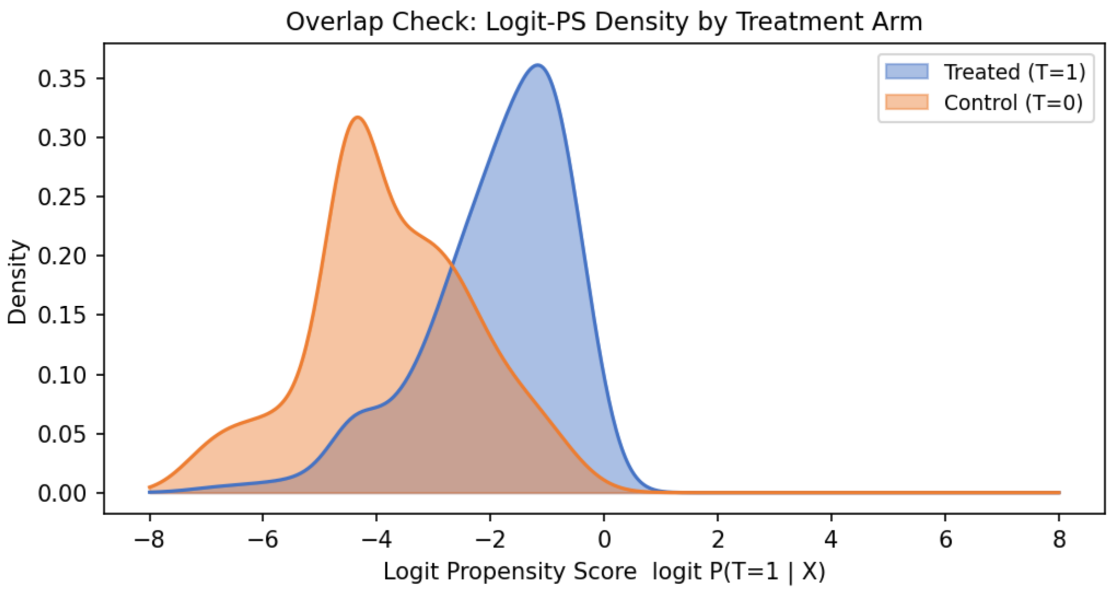

Figure 3: Kernel density estimates of the logit propensity score logit $\hat{e}(\mathbf{X})$ by treatment arm. The treated distribution (blue) peaks at logit ≈ -1.3 and the control distribution (orange) at logit ≈ -4.2. The shaded overlap region represents strata where both treated and control units exist and causal comparisons are well-supported.